Eila Danielle Paley

edpaley@terpmail.umd.edu
Linkedin : linkedin.com/in/edpaley
Github : github.com/edpaley
+1 (301) 957 4309

UMD Undergraduate, Computer Science, Immersive Media Design

Education

University of Maryland, College Park
Bachelor of Science, Computer Science,
Bachelor of Science, Immersive Media Design — *Aug '23 – May '28 (Expected)*
College Park Scholars Living-Learning Program — *Aug '23 – May '25*
Scholars Citation: Environment, Technology, and Economy
GPA: 3.956/4.0 (Overall)

Research Interests

Multi-Agent AI Orchestration, Immersive Social XR & Embodied Agents,
Real-Time Speech AI Pipelines, Empirical Evaluation of Human-Agent Interaction

Publications

Noah Bissell, **Ethan Paley**, Joshua Harrison, Juliano Calil, Myungin Lee, "Sensorium Arc: AI Agent System for Oceanic Data Exploration and Interactive Eco-Art" *Proceedings of the Thirty-Ninth Conference on Neural Information Processing Systems* (NeurIPS 2025), San Diego, CA

Awards

Dean's List, 6 Semesters — *2023–2026*

Research Projects

SociAI (In Progress)
Supervisors : Prof. Huaishu Peng, Prof. Myungin Lee — *July '25 – Present*

- Served as the sole engineer responsible for the end-to-end implementation of the immersive social VR app, integrating RAG architectures and fine-tuned TTS models to replicate specific human personas
- Achieved a conversational pipeline between a Python backend and Unity with under 5 seconds of speech-to-speech latency
- Supporting the design and execution of an upcoming IRB-approved empirical user study

Sensorium Arc: AI Agent System for Oceanic Data Exploration and Interactive Eco-Art
Supervisor : Prof. Myungin Lee — *May '25 – June '26*

- Designed a modular multi-agent system in Unity for a context-grounded conversational pipeline
- Implemented an asynchronous retrieval-augmented generation (RAG) framework using a local vector database and custom query rewriting to ground model responses in curated corpora
- Co-developed the system within a three-member team, translating domain-specific design guidelines from Virtual Planet Technologies and the Center for the Study of the Force Majeure
- Deployed and demonstrated the installation at premier AI venues, including NeurIPS 2025, CVPR 2026, AI+ Expo 2026, and DC Climate Week 2026, validating system stability in high-traffic conditions; delivered an invited technical presentation at ACM SIGGRAPH DAC (SPARKS March 2026)
- *Paper accepted in NeurIPS 2025 (Creative AI Track), one of the premier peer-reviewed conferences in artificial intelligence and machine learning* (Paper)

Leadership & Experience

Teacher's Assistant: Fundamentals of Sculpture
UMD Institute for Advanced Computer Studies, Sandbox Makerspace — *Jan '25 – May '25*

- Led in-class technical workshops on digital fabrication, including 3D printing and laser cutting
- Mentored students individually to brainstorm and bridge creative concepts with physical execution

Engineers Without Borders USA, UMCP Chapter
Supervisor : Prof. Peter Chang — *Sept '24 – Mar '25*

- Coordinated and executed an international monitoring trip to Kiryabicooli, Uganda, managing on-site logistics and stakeholder communications.
- Evaluated structural and electrical performance of two photovoltaic systems to validate long-term sustainability.
- Authored a comprehensive final report and technical documentation to guide future system maintenance and chapter deployment strategies (Report)

Computer Skills

Languages: C#, Python, C, Java, OCaml
Programs: Unity, Git, Adobe Suite, Blender, Fusion 360, Arduino IDE, Orca Slicer